Stress before Distress

Households' Perceived Delinquency Risks and Realized Outcomes

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Motivation

- Households' financial stress is a salient feature of modern capitalist societies
  - $\rightarrow$ Gallup (2024): 28% (48%) of U.S. households worry about a missed (bill) payment
  - $\rightarrow$ FCAC (2025): 56% of Canadians report trouble meeting commitments

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- A gap commonly emerges between perceived stress and realized distress
  - $\rightarrow$ Often a source of public-policy debate: widespread stress, but not in the hard numbers
  - $\rightarrow$ Tempting to dismiss as mere sentiment or survey error, and track only hard numbers

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  - → Tempting to dismiss as mere sentiment or survey error, and track only hard numbers
- This paper, however, takes both measures seriously, and asks:
  - → Why is distress rare while stress prevalent?
  - → What links stress to distress?
  - → Does this gap matter for the macroeconomy?

Preview

Part I — Empirics: how stress maps into distress

1. Stress reasonably co-moves with other expectations in the population
2. Stress predicts distress, but does not always materialize as the latter
3. Empirical measure of the wedge attributable to macro shocks and self-insurance

Part II — A life-cycle model of consumption, borrowing, and delinquency

1. Reconciles broad ex-ante stress with narrow ex-post distress
2. Model view of the wedge: forward-looking self-insurance — anticipating stress, households cut consumption, prepay debt, build buffers

Related literature

- **HA-macro with credit, bankruptcy, and financial stress.** Faccini et al. (2026), Sergeyev, Lian, and Gorodnichenko (2023), Auclert, Dobbie, and Goldsmith-Pinkham (2019), Athreya, Mather, et al. (2019), and Mitman (2016)
  $\rightarrow$ *we add the fear of missing a payment as a distinct precautionary motive*
- **Equilibrium models of consumer default & delinquency.** Athreya, Sanchez, et al. (2018), Athreya, Sánchez, et al. (2015), Chatterjee et al. (2007), and Livshits, MacGee, and Tertilt (2007)
  $\rightarrow$ *we link the delinquency margin to subjective stress and the perceived–realized gap*
- **Subjective expectations in structural models.** Koşar and O'Dea (2023)
  $\rightarrow$ *we compare perceived vs. realized delinquency directly within a life-cycle model*
- **Expectations in consumer credit.** Exler et al. (2024) and D'Acunto, Weber, and Yin (2024)
  $\rightarrow$ *we study perceived delinquency risk and its link to other macro expectations*

Data: measuring stress and distress

- Stress — *perceived* probability of missing a minimum debt payment over the next 3 months
  - → U.S.: NY Fed **Survey of Consumer Expectations** (SCE), monthly, 2013–2024
  - → Canada: Bank of Canada **CSCE**, quarterly, 2014–2024 — identical question
- Distress — *realized* delinquency rates from credit registries
  - → U.S.: FRBNY **Consumer Credit Panel** (Equifax, 5% panel)
  - → Canada: Bank of Canada anonymized **TransUnion** microdata
- Analysis at the **population level**: each measure averaged across households
- Supplementary: **SCF** 2019 (U.S. life cycle), FCAC **Well-Being Monitor** (Canada)

Outline

Empirics: Stress and Distress

- Perceived risk as financial stress
- Does stress realize?
- From stress to distress

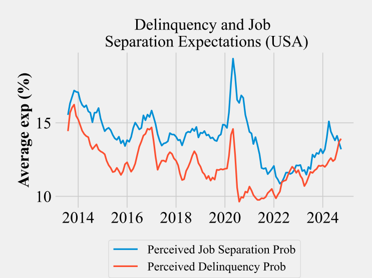
A Life-Cycle Model of Delinquency

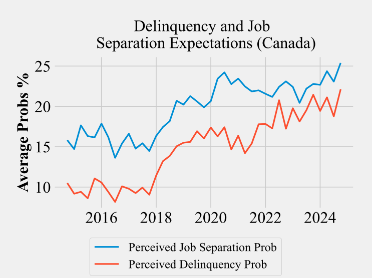
Conclusion

Stress co-moves with unemployment risk

- Perceived delinquency risk co-moves with perceived job-loss risk: U.S. $\rho = 0.57$, Canada $\rho = 0.43$
- Realized job flows track realized delinquency too (pandemic transfers broke the link)

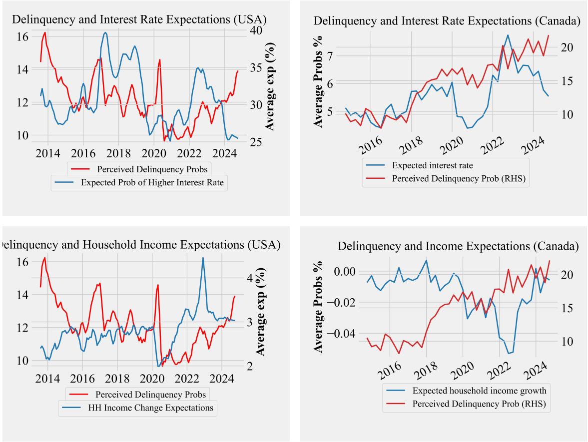
realized job flows





Left: U.S. (SCE, monthly). Right: Canada (CSCE, quarterly).

. . . and with rates, income, and debt burden

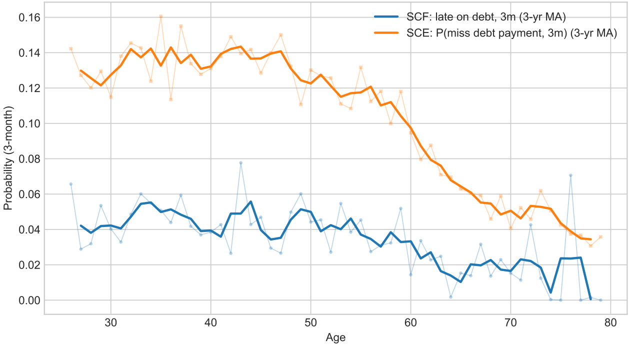


Top: rate exp.; bottom: income exp. Left: U.S.; right: Canada.

- Canada: risk $\uparrow$ with expected *borrowing*-rate hikes ($\rho \approx 0.68$), $\downarrow$ with income growth ($\rho \approx -0.50$)
- U.S.: income $\rho \approx -0.30$; interest $\rho \approx 0.05$ (SCE asks the *saving* rate)
- **DSI:** risk co-moves with aggregate indebtedness (U.S. 0.48, Canada 0.38)

$\Rightarrow$ a credible proxy for financial stress.

Over the life cycle: perceived $\approx 3\times$ realized

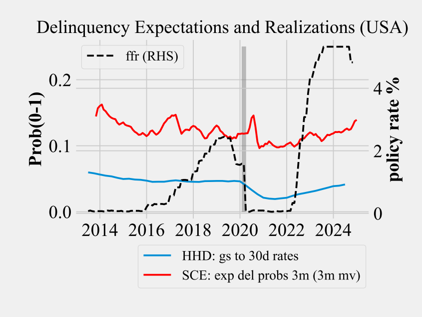


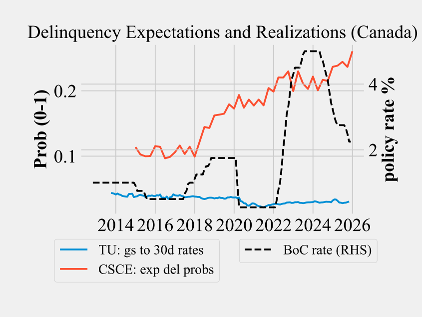
- Actual (SCF 2019): $\sim$4–5% young, falls below 2% by late 60s
- Perceived (SCE): peaks $\sim$14% in the late 30s–40s, declines to $< 5\%$
- Both decline monotonically with age, by roughly the same proportion
- Gap is $\approx$ age-invariant (SCE/SCF ratio 2.84–3.44)

Does stress realize? U.S. yes, Canada no

- Ex-ante perceptions vs. ex-post transition rates: $\rho = \mathbf{0.70}$ (U.S.) but $-\mathbf{0.70}$ (Canada)

$\Rightarrow$ Perceived sits *above* realized in both: stress (broad) vs. distress (narrow); precaution converts stress into *avoided* distress





Left: U.S. (monthly). Right: Canada (quarterly). Source: FRBNY CCP, TransUnion, SCE/CSCE.

From stress to distress: the Stress-to-Distress Ratio (SDR)

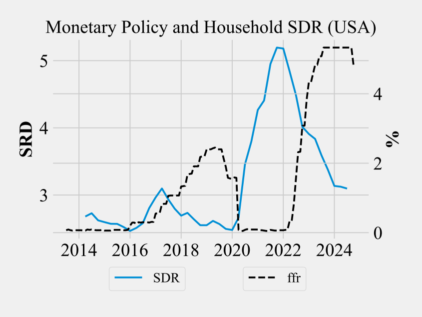
Realized delinquency $=$ perceived probability, scaled by insurance and hit by unexpected shocks:

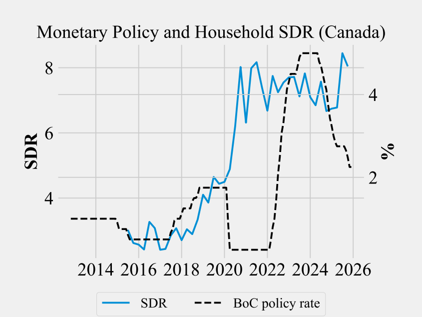
$$\log\left(\frac{\widetilde{\text{Prob}}(D_{i,t+1} \mid \Omega_{i,t})}{\text{Prob}(D_{i,t+1})}\right) = \underbrace{-(\psi_{t+1} + \phi)}_{\textbf{SDR}} \qquad \frac{\text{stress (perceived)}}{\text{distress (realized)}}$$

- **SDR** $=$ combined effect of unexpected macro shocks ψ and insurance ϕ (self- and public)
- Higher SDR $\Rightarrow$ perceived risk far exceeds realized delinquency $\Rightarrow$ stronger resilience

Macro driver: monetary policy rate $\uparrow \rightarrow$ SDR $\downarrow$ (U.S.)

- U.S.: tightening (loosening) is followed by an almost synchronous *fall* (rise) in the SDR — it cycles with policy
- Canada: SDR shows a single *upward trend*, largely insensitive to policy cycles

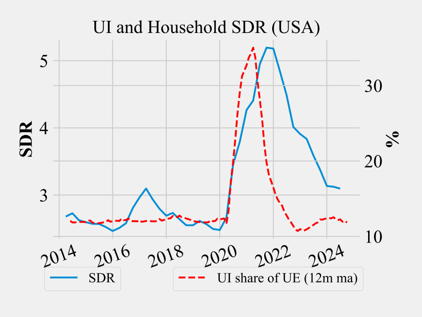


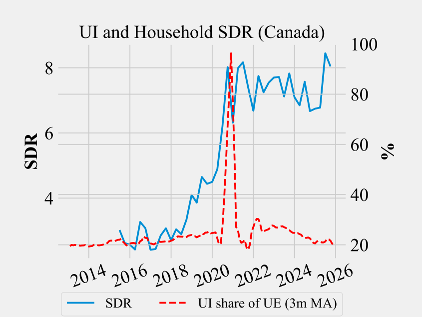


Left: U.S. (monthly). Right: Canada (quarterly). Policy rate on right axis; SDR in exponential form.

Macro driver: more generous UI → SDR ↑ (both)

- SDR is strongly correlated with the UI recipiency rate — the share of the unemployed receiving benefits — in both countries
- **Public insurance** insulates the stress→distress link





Left: U.S. (monthly). Right: Canada (quarterly). Right axis: UI recipiency rate.

Self-insurance: higher stress $\rightarrow$ planned spending cuts

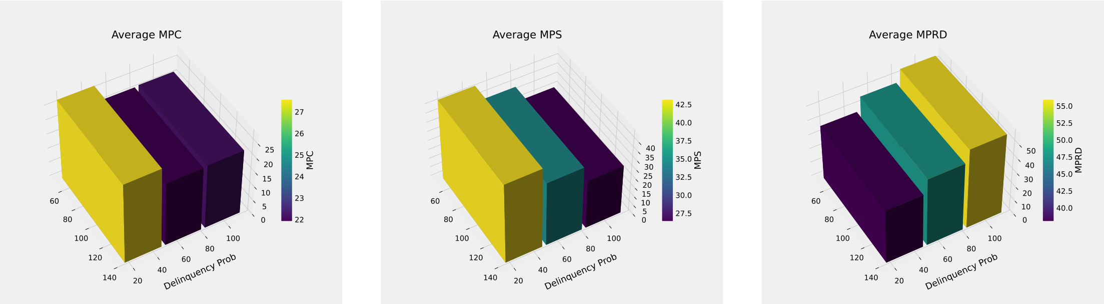
- Higher perceived delinquency risk $\rightarrow$ higher expected spending growth: households plan to cut spending today relative to tomorrow

	(1)	(2)	(3)	(4)
Job Separation Probs			0.03***	
			(0.00)	
Job Finding Probs				0.00**
				(0.00)
Delinquency Probs	**0.04*****	**0.03*****	**0.02*****	**0.04*****
	(0.00)	(0.00)	(0.00)	(0.00)
Expected Wage Growth	0.01			
	(0.01)			
Perceived Wage Risks		0.01***		
		(0.00)		
N	20191	20191	20188	20189
R^2	0.02	0.02	0.03	0.02

Dep. var.: expected spending growth (Canada CSCE). SE clustered by household; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Self-insurance: MPC $\downarrow$, MPRD $\uparrow$

- As perceived delinquency risk rises, households shift marginal resources from consumption toward debt repayment rather than saving



Source: Survey of Consumer Expectations (SCE), U.S. MPC / MPS / MPRD out of a hypothetical windfall, by bins of perceived delinquency risk.

Taking stock: what the empirics tell us

1. Stress is informative. Perceived risk co-moves with other macro expectations (unemployment, rates, income, debt)

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4. Beliefs capture exposure, not the cure. Stress reflects *risk exposure*, but not the insurance that contains distress

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$\Rightarrow$ A life-cycle model with self-insurance reconciles broad stress with narrow distress.

Outline

Empirics: Stress and Distress

A Life-Cycle Model of Delinquency

- Setup and calibration
- Micro behaviors
- Aggregate dynamics

Conclusion

Why a life-cycle model of delinquency?

To reconcile broad ex-ante stress with narrow ex-post distress, we build a life-cycle model of consumption, saving/borrowing, and delinquency:

- Delinquency as a "speed limit," not a crash: costly, but not too costly
  - $\rightarrow$ Existing models jump from repayment to default; this framework fills in between
- Coping capacity under constraints: delinquency as a margin of adjustment when resources are tight
- Forward-lookingness: anticipating stress, households act — adjust consumption, prepay debt, build buffers (self-insurance)

Key Ingredients

- **Income risk:** life cycle profile + permanent/transitory wage shocks + U–E transitions
- **Expenditure shocks:** reduce resources (medical, mortgage rate resets, etc.)
- **Preference shocks:** unobserved shocks influencing pay/not-pay
- **Delinquency costs:** penalty interest, exclusion risk, utility drop
- **Rate differential:** borrowing rate $>$ saving rate $\Rightarrow$ kink at zero wealth
- **Hard borrowing constraint:** exogenous, varies by age
- **Subjective beliefs:** households act on perceived income/employment risk that differs from realized (Wang 2023)

limitations   ▸ model at a glance

Model: Environment

- **Preferences** — CRRA flow utility $+$ warm-glow bequest:

$$\mathbb{E}_0\Big[\sum_{t=0}^{L}\beta^t u(c_t)+\beta^{L+1}\nu\,u(s_{L+1})\Big],\quad u(c)=\frac{c^{1-\gamma}}{1-\gamma}$$

- **Single net balance** s with kinked rates and an exogeneous borrowing limit:

$$\mathcal{R}(s)=\begin{cases}R_s & s\geq 0\\ R & s<0\end{cases}\quad (R>R_s),\qquad s'\geq -\chi$$

- **Income** $y=\Gamma_t\,e^p\,e^\theta\,W_t$: life-cycle profile Γ_t, permanent $p_t=p_{t-1}+\psi_t$, transitory θ_t, employment x (Markov); economy-wide wage W_t
- **Beliefs:** continuation values use perceived risks $\tilde{\mathbb{E}}$; the simulation draws from the actual risks
- **Delinquency costs:** penalty rate $R(1+\eta)$, re-entry prob. λ, flow utility $-\delta$

Model: Recursive Problem

Observe state $(s, p, \theta, \kappa, x)$ and taste shocks; choose *pay/not pay* then c. Primes $=$ next period; $\tilde{\mathbb{E}} =$ perceived.

Pay / save: $\quad v^n(s,\cdot) = \max_{c\geq 0}\; u(c) + \beta\,\tilde{\mathbb{E}}\,W'$

s.t. $\quad y + s\,\mathcal{R}(s) - \kappa = c + s', \quad s' \geq -\chi$

Delinquent: $\quad v^{\mathrm{d}}(s,\cdot) = u(y-\kappa) - \delta + \beta\,\tilde{\mathbb{E}}\big[(1-\lambda)\,v^{\mathrm{d}\prime} + \lambda\,W'\big]$

s.t. $\quad s' = s\,R\,(1+\eta), \quad s' \geq -\chi$

Discrete choice (taste shocks) $\Rightarrow$ log-sum: $W = \sigma\log\big[e^{v^n/\sigma} + e^{v^{\mathrm{d}}/\sigma}\big]$

Savers ($s \geq 0$) always pay; *forced* delinquency when paying is infeasible.

Calibration

Preferences & demographics

β	discount factor	0.94
γ	CRRA / bequest curvature	2
ν	bequest scale	1.0
$L_R,\ L$	retirement / terminal age	65, 85

Prices & credit

$R_s,\ R$	gross save / borrow rate	1.02, 1.08
χ	borrowing limit (/ perm. inc.)	1
W	wage rate (normalized)	1.0

Delinquency

η	penalty markup on R	0.02
λ	re-entry probability	0.90
δ	flow utility penalty	0.7
σ	taste-shock scale	0.8

Income risk — actual vs. perceived

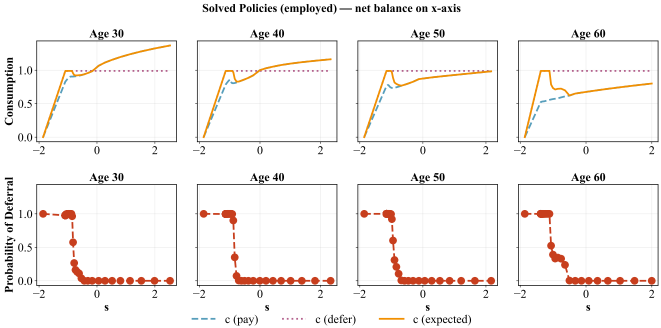
		actual	perceived
σ_ψ	permanent SD	0.10	0.03
σ_θ	transitory SD	0.10	0.03
π_{ue}	job-finding	0.82	0.77
π_{eu}	job-loss	0.04	0.04

Insurance & expenditure

ζ	UI replacement ratio	0.4
ρ^{SS}	SS replacement ratio	0.6
κ	expend. shock $\sim \Gamma(\alpha, \beta)$	(0.02, 0.10)

Γ_t: *hump-shaped life-cycle profile* plot. *Taxes* $\tau = \tau_{SS} = 0$; *levels normalized by permanent income.*

When to pay and when not to pay

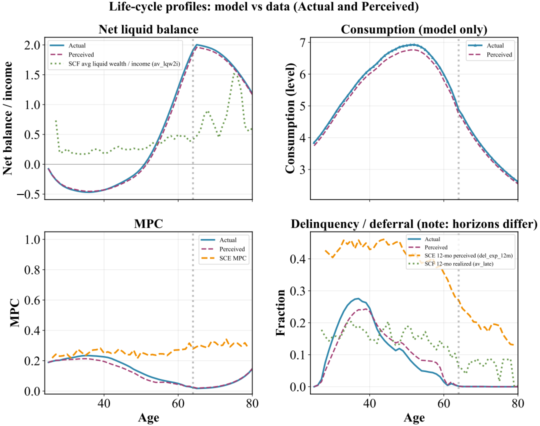


- *Near constraint:* high delinquency probability; *away:* none
- Region shifts with age and income shocks

⇒ Delinquency prob. is a continuous, forward-looking stress measure at every state

- Net balance s alone is a coarse proxy

Life Cycle Profiles



- Hump-shaped delinquency, concentrated at young working ages
- **Actual** (solid): optimize under perceived risk, realized outcomes from the true process
- **Perceived** (dashed): subjective beliefs treated as the truth

Micro Behaviors

Micro-Level Implications

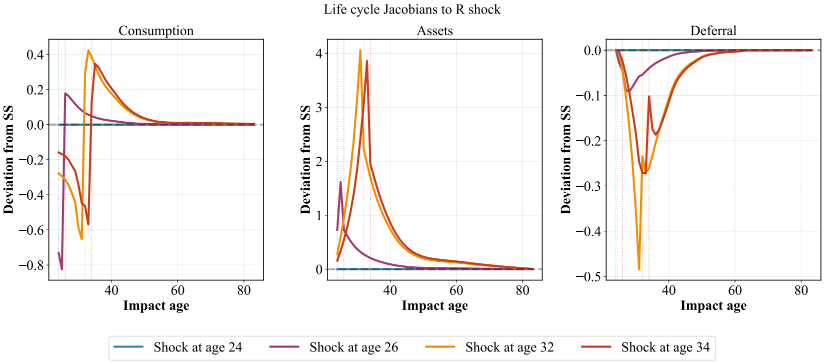
Four implications of the delinquency margin:

- **Non-monotonic MPCs:** the MPC depends on repayment probabilities, unlike the monotonic buffer-stock benchmark details
- **Sharp rise in delinquency near the constraint:** flat at low utilization, hockey-stick rise near the limit details
- **Precautionary prepayment:** penalty scales with debt $\Rightarrow$ accelerate repayment as self-insurance details
- **Voluntary vs. involuntary delinquency:** most — but not all — delinquency is involuntary details

Life-cycle Dynamics via the Jacobian

- **Life-cycle Jacobian:** cohort analogue of the sequence-space Jacobian (Auclert et al. 2021)
- $L \times L$ matrix: response of the cohort path $\{C_j, A_j, D_j\}$ to an anticipated R or W shock at each age
- Anticipation reshapes behavior *backward*; impact & propagation carry it *forward*
- We report: (i) the response to a one-time shock, and (ii) cross-age heterogeneity

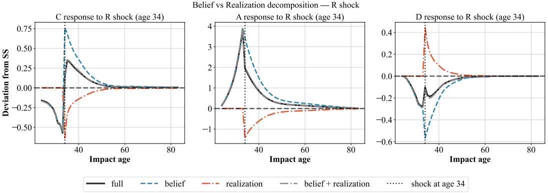
Life-Cycle Response: Interest Rate (R) Shock



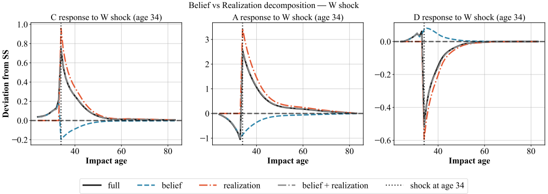
- Anticipating the hike: cut consumption, pay down debt, non-payment falls; reverses at the shock age
- Mirror image for an income/wage gain ▸ W shock

Jacobian Decomposition at a Given Age (Age 34)

Interest rate shock



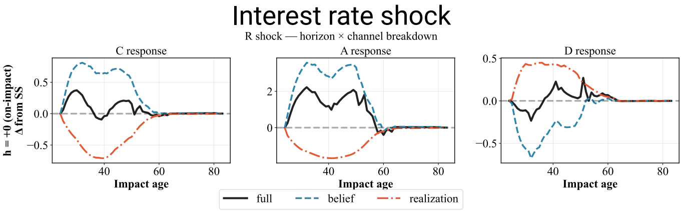
Income shock

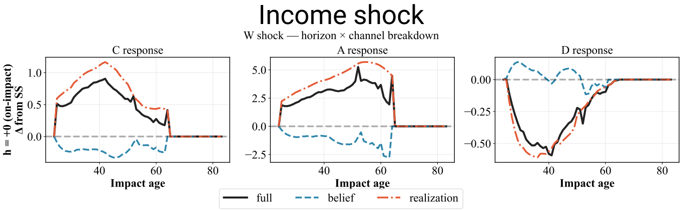


- Realization channel: nonzero only from the shock age forward
- Belief channel: nonzero at *all* ages — anticipatory paydown before, altered states after
- Belief response mitigates the realization channel: not all stress becomes distress

$\Rightarrow$ SDR $\approx$ belief / full

Belief Channel Mitigates Stress Impact of Shocks



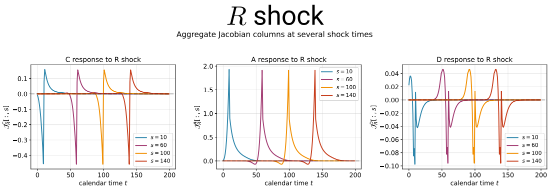


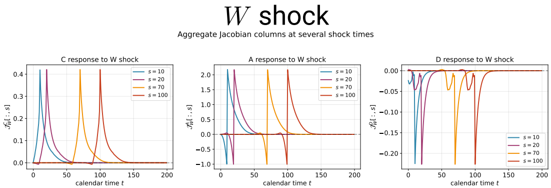
- Anticipation absorbs impact-period stress; total consumption/delinquency responses can flip sign at impact
- Stress-to-distress ratio differs across ages; can be both positive and negative

▸ across horizons

Aggregate Dynamics

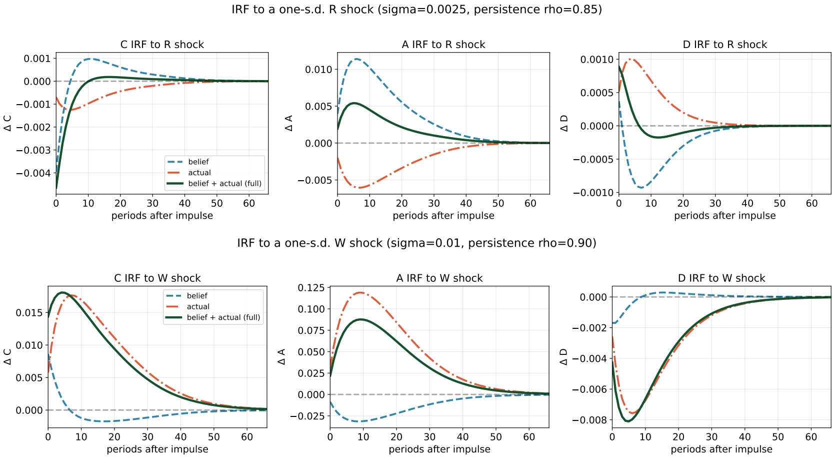
From Life-Cycle to Aggregate Jacobian





- Each column $=$ population-weighted average of the life-cycle Jacobian (cohort weights from the stationary age distribution)
- The belief channel populates the off-diagonal mass — anticipated shocks bite away from impact

Impulse Responses to Persistent Aggregate Shocks



Each row: belief vs. realization channels and their sum, against the full response.

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- In U.S. and Canadian data, stress is widespread but distress is rare
- Perceived risk is a forward-looking gauge of stress: co-moves with job, rate, income, debt risks; falls with age
- Stress→distress tracks the cycle, buffered by public and self-insurance and milder-than-feared shocks

Conclusion

- In U.S. and Canadian data, stress is widespread but distress is rare
- Perceived risk is a forward-looking gauge of stress: co-moves with job, rate, income, debt risks; falls with age
- Stress$\to$distress tracks the cycle, buffered by public and self-insurance and milder-than-feared shocks
- A life-cycle model with a delinquency choice delivers a *continuous* model-implied stress measure, mapped directly to the data
- This model reconciles the stress–distress wedge arising from self-insurance and subjective beliefs

Limitations / What the Model Does Not Do

- **No mortgage / secured debt default** — unsecured credit only
  - $\rightarrow$ *But:* can still think about a mortgager's problem facing payment shocks
- **No persistent credit score impacts** — exclusion risk is stylized
  - $\rightarrow$ *But:* the utility cost captures it somewhat
- **Single asset** — no portfolio choice or liquid/illiquid split
  - $\rightarrow$ *But:* implicitly assumes illiquid asset adjustment is costly
- **No co-holding** — no simultaneous saving and borrowing
  - $\rightarrow$ *But:* still captures revolving behaviors and non-monotonic MPCs
- **No costly bankruptcy** — no foreclosure, fresh start, or permanent exclusion
  - $\rightarrow$ *But:* captured by the hard borrowing constraint

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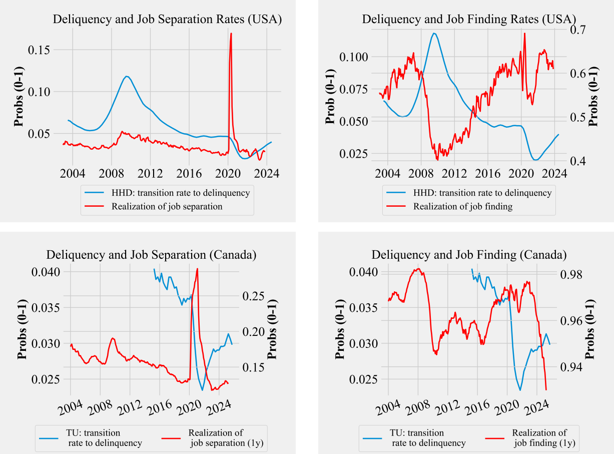
Realized job flows track realized delinquency



Realized job-separation (left) & job-finding (right) vs. realized delinquency. Top: U.S., bottom: Canada.

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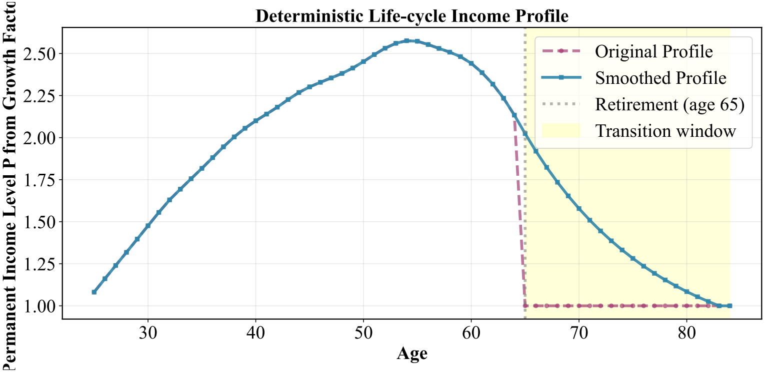
Deterministic Life Cycle Income Profile



- Hump-shaped profile: peaks mid-50s, declines into retirement
- Smoothed transition around retirement (age 65) for numerical stability

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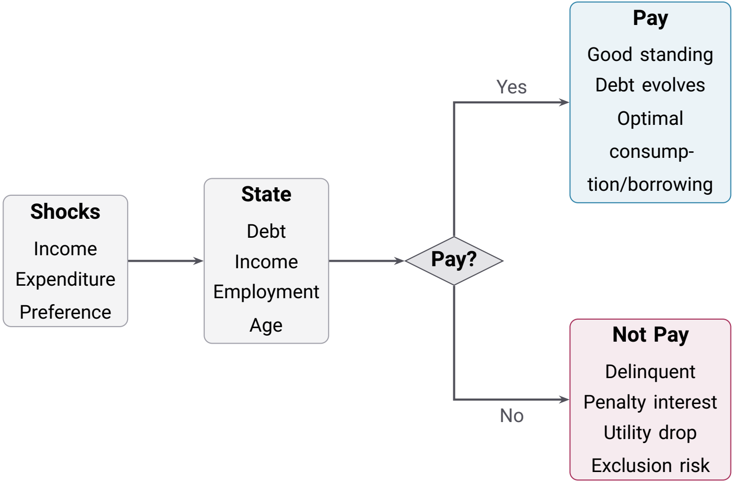
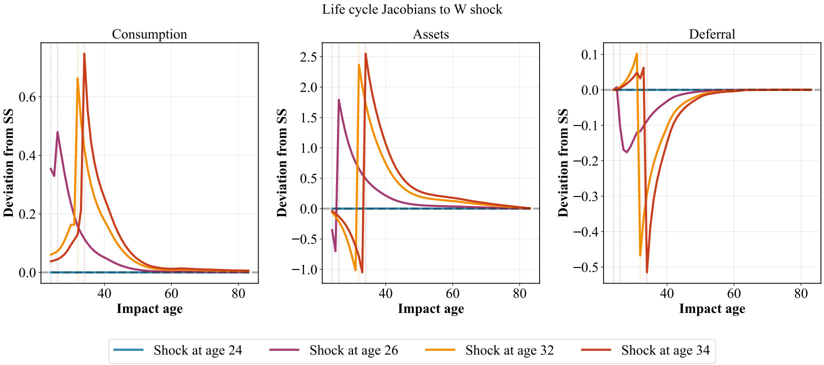
Model at a Glance



Repeats over the life cycle: young → middle → old

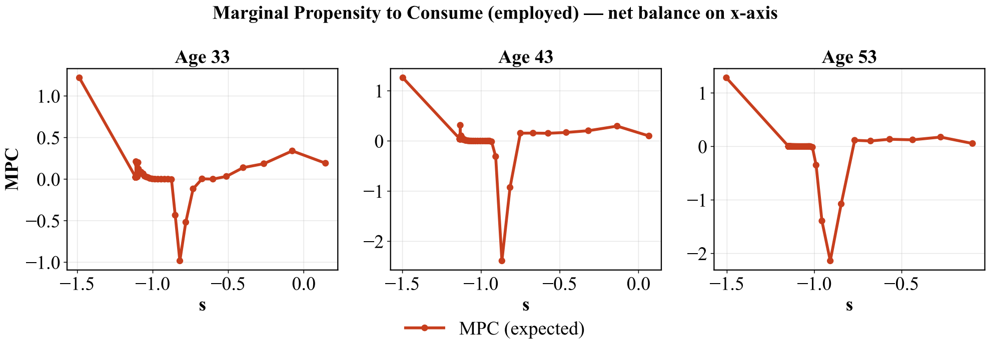
Life-Cycle Response: Income/Wage (W) Shock



- Mirror image: anticipating the gain, households borrow ahead, non-payment rises first, then subsides

back

Implication: Non-Monotonic MPCs



- MPC depends on repayment probabilities
- Contrast: buffer-stock model $\Rightarrow$ monotonic MPC

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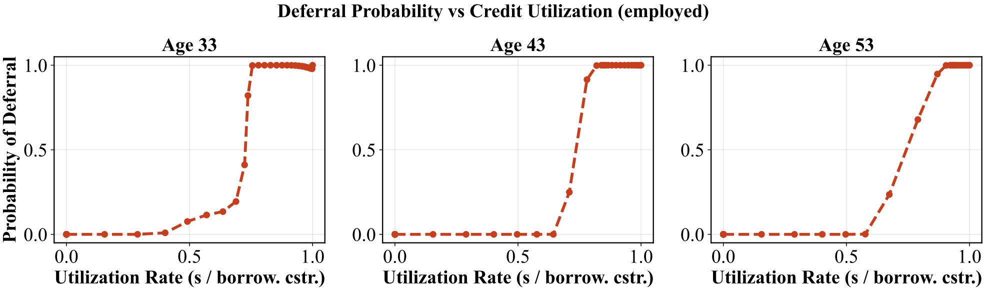
Implication: Sharp Rise in Delinquency near the Constraint



- Flat at low utilization, hockey-stick rise near limit
- Low utilization: has slack absorbs shocks, but can still fall into delinquency

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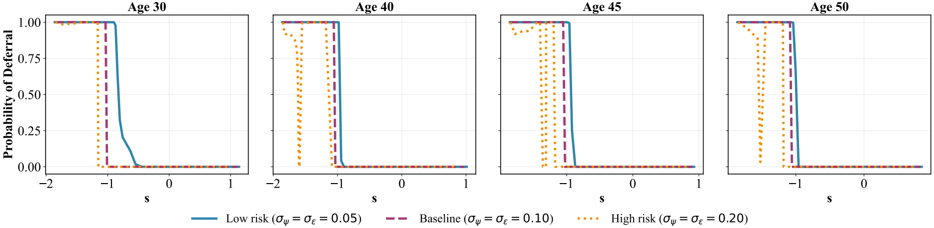
Implication: Precautionary Prepayment



- Penalty scales with debt $\Rightarrow$ higher expected risk $\Rightarrow$ accelerate repayment
- Self-insurance, distinct from precautionary *saving*; stronger with κ/unemployment risk

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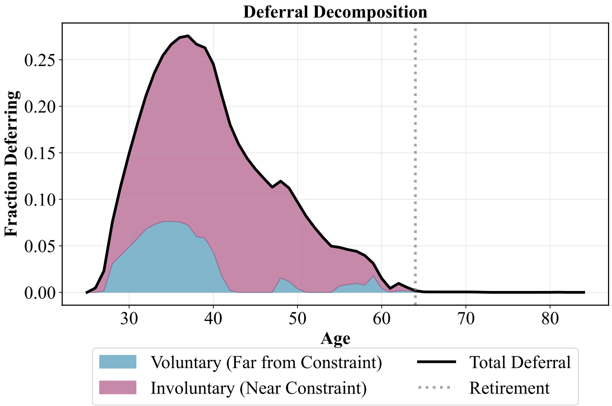
Implication: Voluntary vs. Involuntary Delinquency



- **Voluntary:** could repay, chooses not to, "splurge" consumption
- **Involuntary:** cannot repay — insufficient resources
- Key result: most but not all delinquency is involuntary

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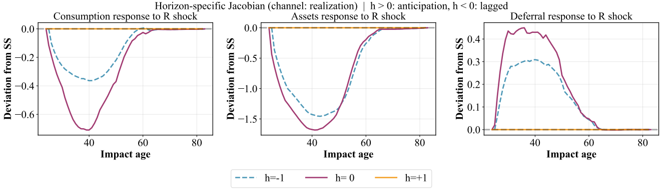
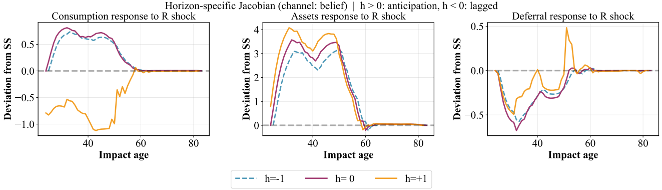
Heterogeneous Responses across Horizons: R Shock

Realization channel



Belief channel



- Realization channel is zero before the shock ($h = +1$); belief channel is active at all horizons
- Belief response flips sign for past vs. future shocks — precaution builds, then unwinds

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